

Notes: ‘n’ represents the dimension of the corresponding problem. ‘NI’ represents the total number of iterations. ‘NF’ represents the number of the function evaluation. ‘T’ represents the running time of the algorithm. ‘ $\|g^*\|$ ’ represents the gradient value for the solution. The values of ‘NI’, ‘NF’, ‘T’ and ‘ $\|g^*\|$ ’ are set as ‘NaN’ if and only if the number of iterations is greater than 3000.

Table 1: Numerical results for Test 2

Problem/n	SHCG-V3 NI/NF/T/ $\ g^*\ $	SHCG-V4 NI/NF/T/ $\ g^*\ $	HZ NI/NF/T/ $\ g^*\ $	FhDY NI/NF/T/ $\ g^*\ $	MP+ NI/NF/T/ $\ g^*\ $	SMABFGS NI/NF/T/ $\ g^*\ $
dixmaana/3000	21/81/0.064/9.15e-07	22/83/0.058/6.63e-07	24/86/0.091/3.43e-07	19/83/0.057/7.02e-08	18/80/0.055/6.21e-07	45/96/0.069/1.69e-07
dixmaana/30000	20/86/0.475/9.75e-07	25/94/0.521/7.64e-07	23/93/0.500/3.73e-07	19/86/0.469/3.86e-07	24/92/0.507/9.33e-07	82/135/0.770/8.66e-07
dixmaana/600000	14/84/8.891/6.03e-07	20/96/10.190/4.88e-07	24/103/11.233/8.47e-07	17/90/9.539/6.14e-07	17/90/9.555/3.33e-07	92/142/15.207/3.12e-07
dixmaanb/6000	18/84/0.107/6.07e-07	16/80/0.098/7.42e-07	18/83/0.143/7.90e-07	16/82/0.102/2.89e-07	21/88/0.110/2.72e-07	49/101/0.127/5.24e-07
dixmaanb/15000	20/87/0.279/8.58e-08	23/91/0.275/9.48e-08	25/93/0.281/3.56e-07	22/90/0.263/4.55e-09	21/86/0.255/9.31e-07	72/125/0.378/3.71e-07
dixmaanb/60000	23/93/0.942/6.05e-07	17/83/0.836/7.77e-07	20/89/0.907/5.78e-07	23/98/0.993/9.52e-07	20/88/0.893/4.08e-07	75/131/1.342/3.00e-08
dixmaanc/3000	18/81/0.051/9.54e-07	14/79/0.051/8.38e-07	23/86/0.076/9.99e-07	19/84/0.053/5.05e-07	23/87/0.055/2.89e-07	40/86/0.055/2.22e-07
dixmaanc/150000	21/92/2.495/9.06e-07	24/97/2.642/4.42e-07	28/106/2.855/1.74e-07	20/95/2.573/6.21e-07	24/97/2.618/3.42e-07	90/146/3.979/6.97e-07
dixmaand/300000	19/91/4.834/6.66e-07	29/104/5.532/8.72e-07	26/101/5.455/5.62e-07	17/80/4.280/8.83e-07	22/97/5.197/6.38e-07	74/128/6.954/5.07e-07
dixmaane/1500	315/399/0.127/6.24e-07	357/419/0.133/7.91e-07	390/821/0.291/8.79e-07	308/589/0.206/9.87e-07	306/507/0.175/9.10e-07	382/430/0.151/5.60e-07
dixmaane/3000	478/583/0.373/8.63e-07	477/541/0.350/9.94e-07	580/1265/0.820/5.02e-07	433/802/0.507/9.47e-07	473/785/0.497/7.43e-07	665/726/0.472/4.64e-07
dixmaane/15000	825/1039/10.317/7.91e-07	997/1064/10.049/9.81e-07	1134/2403/15.066/7.77e-07	753/1464/12.816/9.06e-07	1132/1838/15.284/7.74e-07	1443/1524/15.210/9.38e-07
dixmaanf/1500	265/358/0.114/8.35e-07	363/427/0.148/9.80e-07	415/882/0.634/7.97e-07	246/483/0.310/9.56e-07	226/395/0.219/5.36e-07	401/450/0.172/9.78e-07
dixmaanf/3000	359/466/0.303/7.27e-07	364/428/0.289/9.27e-07	283/621/0.397/6.49e-07	287/531/0.354/8.60e-07	271/442/0.342/8.18e-07	698/750/0.491/9.74e-07
dixmaanf/30000	783/996/17.605/6.94e-07	1627/1723/32.409/7.51e-07	1014/2182/35.624/9.24e-07	834/1619/23.493/6.56e-07	1010/1653/27.642/9.93e-07	NaN/NaN/NaN/NaN
dixmaang/300	158/233/0.036/8.75e-07	279/338/0.060/9.76e-07	162/353/0.160/9.46e-07	115/237/0.040/6.40e-07	98/186/0.030/8.38e-07	139/188/0.033/9.94e-07
dixmaang/1500	332/432/0.241/8.96e-07	502/571/0.321/9.00e-07	282/626/0.345/8.26e-07	266/520/0.288/9.22e-07	215/377/0.201/9.46e-07	327/381/0.209/7.25e-07
dixmaang/15000	1648/1986/18.911/8.60e-07	NaN/NaN/NaN/NaN	817/1747/18.272/8.45e-07	NaN/NaN/NaN/NaN	1664/2791/22.236/9.75e-07	1016/1079/11.735/7.24e-07
dixmaanh/15000	719/889/8.606/9.80e-07	932/1017/8.273/9.02e-07	799/1776/18.494/9.56e-07	520/995/10.355/6.81e-07	703/1136/12.830/8.23e-07	NaN/NaN/NaN/NaN
dixmaanh/21000	1984/4380/17.636/8.67e-07	1130/1234/5.017/9.71e-07	NaN/NaN/NaN/NaN	851/1615/23.001/9.00e-07	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
dixmaanh/30000	1095/1358/7.416/8.46e-07	1157/1249/6.853/9.56e-07	1187/2548/13.992/1.00e-06	921/1738/9.456/9.95e-07	859/1401/7.666/7.94e-07	NaN/NaN/NaN/NaN
dixmaani/30	203/266/0.007/9.81e-07	257/324/0.008/9.08e-07	289/611/0.038/8.04e-07	202/398/0.010/9.14e-07	197/332/0.008/9.60e-07	257/305/0.008/8.33e-07
dixmaani/90	632/746/0.026/8.92e-07	812/888/0.030/9.46e-07	660/1388/0.055/9.99e-07	477/904/0.037/8.24e-07	714/1191/0.045/8.00e-07	948/1015/0.037/9.27e-07
dixmaani/210	1242/1452/0.084/8.82e-07	1224/1284/0.074/9.70e-07	1803/3855/0.225/8.49e-07	1185/2267/0.132/9.99e-07	1714/2830/0.163/8.71e-07	1870/1968/0.115/8.96e-07
dixmaanj/600	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	1022/2235/0.386/9.35e-07	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
dixmaanj/6000	745/925/1.153/6.73e-07	NaN/NaN/NaN/NaN	917/2000/2.438/9.00e-07	592/1081/1.320/8.21e-07	887/1502/1.850/8.84e-07	NaN/NaN/NaN/NaN
dixmaanl/300	1305/1572/0.122/7.68e-07	NaN/NaN/NaN/NaN	998/2157/0.231/9.90e-07	868/1647/0.141/9.20e-07	1375/2287/0.173/8.98e-07	NaN/NaN/NaN/NaN
dixmaanl/21000	952/1181/4.809/9.08e-07	NaN/NaN/NaN/NaN	801/1750/6.885/6.44e-07	1331/2530/10.062/9.23e-07	1128/1900/7.687/9.40e-07	NaN/NaN/NaN/NaN
dixmaanl/30000	1141/1394/27.864/9.84e-07	1702/1775/33.391/9.78e-07	1121/2420/13.169/9.29e-07	674/1299/7.076/8.20e-07	1180/1970/10.953/8.15e-07	NaN/NaN/NaN/NaN
dixon3dq/50	488/623/0.036/9.40e-07	592/670/0.038/8.22e-07	613/1298/0.117/9.32e-07	616/1166/0.086/8.84e-07	553/764/0.059/9.40e-07	493/537/0.027/8.84e-07
dixon3dq/100	936/1144/0.043/8.27e-07	779/841/0.036/8.10e-07	1920/4108/0.147/9.41e-07	821/1576/0.050/9.16e-07	1242/1712/0.070/8.99e-07	1158/1235/0.054/9.27e-07
dixon3dq/150	1815/2184/0.132/8.29e-07	1378/1432/0.091/8.41e-07	1834/3969/0.176/9.57e-07	1223/2322/0.121/5.56e-07	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
dqdrtic/10000	77/228/0.029/5.19e-07	111/262/0.040/5.89e-07	97/275/0.077/3.99e-07	74/222/0.037/9.40e-07	86/199/0.039/6.74e-07	142/209/0.072/2.65e-07
dqdrtic/50000	100/294/0.178/7.26e-07	119/293/0.181/9.05e-07	113/306/0.176/5.12e-07	82/210/0.115/5.14e-07	86/230/0.144/9.52e-07	193/263/0.253/3.59e-07
dqdrtic/150000	76/221/0.950/3.47e-07	117/312/1.116/6.64e-07	116/334/0.865/3.46e-07	66/241/0.914/6.97e-07	70/203/0.783/6.76e-07	138/213/0.761/7.72e-07
dqrtic/5000	56/168/0.260/6.27e-07	61/173/0.284/6.38e-07	51/169/0.320/4.54e-07	41/156/0.241/4.19e-07	79/360/0.587/6.14e-07	96/185/0.317/3.43e-07
edensch/500	44/195/0.036/8.63e-07	53/150/0.065/7.67e-07	43/131/0.068/6.19e-07	68/474/0.209/3.98e-07	50/125/0.032/9.90e-07	144/941/0.178/9.60e-07
edensch/20000	78/502/5.694/4.01e-07	66/229/2.618/8.00e-07	66/385/4.128/9.49e-07	51/269/3.125/5.85e-07	43/157/1.819/7.82e-07	394/2923/32.142/9.95e-07
edensch/120000	69/375/17.580/6.99e-07	121/751/29.194/5.46e-07	61/230/11.235/8.70e-07	125/965/49.940/2.12e-07	146/1239/56.663/4.71e-07	882/7254/359.194/9.96e-07
eg2/100	303/1422/0.072/7.68e-07	171/560/0.024/5.65e-07	1972/5805/0.353/8.98e-07	538/1585/0.097/4.46e-07	166/591/0.035/8.56e-07	1086/1250/0.080/8.55e-07
eg2/200	343/1322/0.099/8.24e-07	431/2338/0.211/8.53e-07	NaN/NaN/NaN/NaN	1122/5148/0.437/8.30e-07	980/8893/0.603/8.93e-07	NaN/NaN/NaN/NaN
fletcher/500	115/887/0.018/7.33e-07	254/1904/0.065/3.46e-07	102/650/0.093/4.21e-07	69/373/0.028/4.37e-07	83/583/0.033/8.79e-07	NaN/NaN/NaN/NaN
fletcher/7000	260/2188/0.261/7.02e-07	446/4053/0.522/1.11e-07	152/946/0.154/5.93e-07	391/3583/0.430/9.27e-07	89/555/0.080/8.86e-07	NaN/NaN/NaN/NaN
freuroth/50	181/550/0.018/8.05e-07	404/1985/0.077/7.64e-07	398/2571/0.118/9.13e-07	236/1585/0.052/9.83e-07	221/906/0.030/9.24e-07	NaN/NaN/NaN/NaN
freuroth/500	1087/10723/0.632/9.94e-07	903/8818/0.453/9.75e-07	597/5074/0.296/1.51e-07	NaN/NaN/NaN/NaN	1746/18090/1.069/5.12e-07	NaN/NaN/NaN/NaN
freuroth/5000	923/9512/1.887/7.43e-07	NaN/NaN/NaN/NaN	1897/19001/3.353/8.98e-07	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
genrose/10000	290/462/0.085/9.99e-07	380/519/0.092/9.64e-07	310/672/0.210/9.51e-07	225/473/0.089/7.75e-07	230/368/0.073/9.69e-07	397/475/0.188/5.53e-07
himmelbg/1000	2/13/0.001/7.55e-30	2/13/0.001/7.55e-30	2/13/0.025/5.64e-30	2/13/0.002/7.57e-30	2/13/0.001/1.99e-31	8/12/0.002/1.59e-42
himmelbg/70000	2/15/0.105/8.37e-43	2/15/0.114/8.37e-43	2/15/0.053/5.51e-43	2/15/0.086/8.41e-43	2/15/0.097/7.95e-47	8/15/0.118/9.38e-10
himmelbg/600000	3/16/0.790/1.30e-26	3/16/0.715/1.61e-26	3/16/0.485/1.08e-26	3/16/0.523/8.54e-27	3/14/0.558/1.35e-62	11/19/0.779/8.27e-20
liarwhd/10000	128/456/0.056/3.94e-07	124/455/0.057/8.36e-07	285/875/0.276/6.61e-07	176/628/0.085/5.48e-07	98/421/0.056/9.04e-07	853/1117/0.391/6.11e-07
liarwhd/30000	164/562/0.274/5.06e-07	136/577/0.260/9.63e-07	299/974/0.419/4.86e-07	201/732/0.284/2.54e-07	147/584/0.262/6.53e-07	716/994/0.785/2.91e-07
liarwhd/120000	164/651/0.721/1.51e-07	133/594/0.646/5.64e-07	431/1386/1.425/9.89e-07	315/1075/1.129/4.67e-07	148/665/0.762/9.07e-07	679/1018/1.786/8.45e-07
penalty1/100	19/109/0.007/8.13e-07	19/109/0.006/9.29e-07	18/108/0.038/8.27e-07	18/107/0.006/2.82e-07	19/109/0.006/9.20e-07	21/68/0.004/4.28e-07
penalty1/10000	20/92/13.391/4.52e-07	20/92/13.316/1.83e-07	18/87/11.246/4.89e-07	20/91/12.285/8.17e-07	20/92/12.432/4.55e-07	113/180/23.612/3.87e-07
quartc/5000	56/168/0.263/6.27e-07	61/173/0.267/6.38e-07	51/169/0.309/4.54e-07	41/156/0.288/4.19e-07	79/360/0.600/6.14e-07	96/185/0.513/3.43e-07
tridia/100	251/341/0.016/8.82e-07	295/364/0.018/9.16e-07	494/1058/0.056/7.94e-07	353/664/0.024/6.72e-07	369/497/0.018/9.70e-07	365/426/0.034/4.46e-07
tridia/300	623/768/0.041/9.58e-07	512/585/0.022/9.22e-07	1080/2297/0.102/8.66e-07	1789/1519/0.078/6.12e-07	756/986/0.059/9.06e-07	797/871/0.037/9.36e-07
tridia/1500	1960/2355/0.218/9.73e-07	1482/1567/0.179/9.96e-07	NaN/NaN/NaN/NaN	544/2944/0.170/6.80e-07	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
woods/5000	158/434/0.047/8.84e-07	300/578/0.060/4.16e-07	371/920/0.152/5.45e-07	263/640/0.066/7.03e-07	147/439/0.048/8.27e-07	262/336/0.095/3.07e-07
woods/300000	210/530/4.086/8.38e-07	300/650/4.980/2.03e-07	475/1197/8.148/9.22e-07	376/910/6.810/8.20e-07	187/477/3.695/5.19e-07	324/401/4.018/8.73e-07
woods/1000000	218/648/13.120/5.72e-07	376/750/15.622/6.87e-07	660/1661/34.254/9.34e-07	281/702/15.162/7.37e-07	196/487/11.400/7.11e-07	396/479/13.221/6.24e-07
bdexp/1000	2/11/0.003/2.99e-133	2/11/0.003/2.99e-133	2/11/0.053/4.48e-139	2/11/0.007/3.25e-133	2/11/0.006/4.46e-133	13/14/0.009/1.41e-11
bdexp/10000	2/17/0.016/2.62e-61	2/17/0.018/2.62e-61	2/17/0.040/1.25e-63	2/17/0.024/2.63e-61	2/17/0.020/2.68e-61	37/39/0.077/0.00e+00
bdexp/100000	2/12/0.205/4.57e-106	2/12/0.225/4.57e-106	2/12/0.276/4.45e-106	2/12/0.279/4.57e-106	2/12/0.244/4.58e-106	30/37/0.809/0.00e+00
exdenschnf/10000	40/113/0.018/8.83e-07	48/124/0.023/4.67e-07	37/115/0.056/3.69e-07	25/99/0.016/4.94e-08	31/106/0.019/9.01e-07	55/118/0.036/4.22e-07
exdenschnb/3000	26/89/0.005/9.86e-07	47/111/0.007/1.10e-07	25/87/0.019/5.00e-07	19/84/0.007/8.43e-09	27/88/0.007/7.04e-07	50/98/0.011/8.07e-07
exdenschnb/5000000	49/125/9.941/3.21e-07	36/111/7.722/4.17e-07	33/110/8.888/3.55e-08	32/107/7.914/4.32e-07	34/108/9.009/9.39e-07	104/164/17.828/4.03e-07
genquartic/50000	25/93/0.092/3.19e-07	29/97/0.087/8.47e-07	23/92/0.113/3.00e-07	24/116/0.099/3.82e-07	26/96/0.093/3.24e-07	63/117/0.149/9.80e-07
genquartic/100000	21/86/0.097/7.90e-07	24/93/0.105/9.72e-07	27/97/0.143/8.69e-07	19/96/0.116/3.50e-07	28/96/0.112/7.67e-07	75/128/0.206/3.92e-07
genquartic/200000	0/0/0.002/0.00e+00	0/0/0.002/0.00e+00	0/0/0.002/0.00e+00	0/0/0.002/0.00e+00		